

迈克尔逊干涉 (1)

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2025 年 10 月 22 日

1 数据

δn	0	30	60	90
x/mm	46.21255	46.20297	46.19391	46.18468

2 计算

氦氖激光器的波长为

$$\lambda_{\text{标准}} = 632.8nm$$

直接计算平均值

$$\lambda = \frac{2\Delta d}{\Delta n} = \frac{2 \times (\Delta d_1 + \Delta d_2 + \Delta d_3)}{30 \times 3} = 619.33mm$$

相对误差

$$\frac{|619.33mm - 632.8mm|}{632.8mm} \times 100\% = 2.13\%$$

逐差法

$$\lambda = \frac{n_3 + n_4 - n_1 - n_2}{(2 \times 30)^2} = 615.50mm$$

相对误差

$$\frac{|615.5m - 632.8mm|}{632.8mm} \times 100\% = 2.73\%$$

3 误差分析

$$u_A = \sqrt{\frac{\sum_{i=1}^N (A_i - \bar{A})^2}{N(N-1)}} = \sqrt{\frac{\sum_{i=1}^3 (\Delta d_i - \bar{\Delta d})^2}{3 \times (3-1)}} = 153.7nm$$

$$u_B = \frac{1 \times 10^{-7}}{\sqrt{3}}m = 57.74nm$$

$$u = \sqrt{u_A^2 + u_B^2} = 163.605nm$$

$$\frac{\partial \lambda}{\partial \Delta d_i} = \frac{1}{45}$$

$$U = \sqrt{\sum_{i=1}^3 \frac{\partial \lambda}{\partial \Delta d_i}} \times u = 3.3025769321445257nm$$

则

$$\lambda = (619.33 \pm 3.303)nm$$